

Design, implementation, and reflections on delivering machine learning workshops for medical microbiology and infection science

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1.4 Keywords

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2. Abstract

Increasingly, machine learning (ML) and artificial intelligence (AI) methods are being applied in medical microbiology and infection science. With this comes the challenge of educating professionals in these domains on the fundamental theory and methods of ML and AI in a way that is time-efficient and grounded in practical application. This article outlines the design, implementation, and reflections on delivering “Using Data Science and Machine Learning for Infection Science: A Hands-On Introduction”, a recurring one-day workshop created to introduce data science and ML concepts to infection science and medical

microbiology students and professionals. The workshop provides participants with the opportunity for experiential learning using *Orange*, a no-code, open-source, and free-to-use data mining software application, where participants are exposed to the fundamentals of the ML lifecycle. By the conclusion of the one-day workshop, participants gain experience developing end-to-end analytical pipelines for tabular datasets, with a specific example focused on blood culture outcome prediction. The workshop emphasises transparency, reproducibility, and open science, promoting critical awareness of how data science can be applied within participants' own domain specialty. This article details the curriculum design, practical implementation, and delivery of the workshop. Additionally, we reflect on the lessons learned and directions for future iterations of the workshop.

3. Impact statement

Artificial intelligence (AI) and machine learning (ML) are increasingly used in medical microbiology and infection science, but many professionals have limited time or opportunity to learn these methods in a practical, accessible way. This article describes the development and delivery of a one-day workshop designed to introduce students and professionals to the key ideas behind data science and machine learning. The workshop uses Orange, a free, open-source, no-code software tool, so participants can focus on understanding the machine learning process without needing advanced programming skills.

Through hands-on activities, participants learn how to build simple analytical workflows using real-world style tabular data. The workshop also highlights important principles such as transparency, reproducibility, and open science, helping participants think critically about how these methods could be used responsibly in their own work.

By sharing the workshop design, teaching approach, and lessons learned, this article provides a practical model for building machine learning education within medical microbiology and infection science. It supports the development of a more confident and informed workforce able to engage with emerging AI, ML, and data science tools.

4. Data summary

The authors confirm that no data was generated or reused in the process of producing this paper.

63 5. Introduction

64 Digital transformation across healthcare has led to an unprecedented accumulation of
65 laboratory, clinical, and infection-related data. In the context of clinical microbiology and
66 infection science, this data holds the potential to support a variety of areas, including
67 diagnostic decision-making, antimicrobial resistance monitoring, and process improvement.
68 Alongside the increasing volume of data, machine learning (ML) and artificial intelligence
69 (AI) methods are being applied more widely to detect patterns in datasets and develop
70 predictive models [1]. However, many infection science professionals lack formal training in
71 computational methods, which limits the translation of raw data into actionable insights and
72 applications. Furthermore, as research on the application of these methods increases, there
73 is a need for infection science and clinical microbiology professionals to have the knowledge
74 to support peer-review and aid in translating applications into practice. In the literature,
75 there are previously published tutorials and workshops for healthcare professionals and for
76 those without ML domain expertise [2, 3]. However, to the best of our knowledge, there are
77 no published papers discussing in-person data science and ML workshops tailored to
78 infection science.

79 To address this gap in training amongst those working and/or studying in the clinical
80 microbiology and infection science domain, authors Benjamin McFadden (BM) and Timothy
81 Inglis (TI) designed the *“Using Data Science and Machine Learning for Infection Science: A
82 Hands-On Introduction”* one-day workshops. Since 2024, the workshop has been delivered
83 at the ECCMID conference in April 2024 (Barcelona, Spain), and the ESCMID global
84 conference in April 2025 (Vienna, Austria) and April 2026 (Munich, Germany) with BM and TI
85 both present to deliver these workshops. The workshop content was also delivered by BM at
86 the University of Zurich Medical Microbiology Institute in June 2025, and again in April 2026.
87 Prior to 2024, an earlier version of the workshop was created by TI and delivered in 2019 at
88 the ECCMID conference (Amsterdam, Netherlands). The planned sequel at ECCMID 2020 was
89 postponed due to the COVID-19 pandemic. Over its lifetime, the workshop has attracted
90 participants from diverse worldwide settings including hospitals, research institutes, and
91 postgraduate programs. The aim of these workshops is to teach the principles and practice
92 of reproducible ML projects from an end-to-end perspective. This means taking a holistic
93 view, starting with data preparation, and progressively working through data processing and
94 cleaning, ML modelling, and ML model evaluation. The teaching of these concepts is
95 achieved using the *Orange* data mining software [4], an open-source, no-code tool that
96 supports a range of analytical tasks, including data cleaning and transformation,
97 visualisation, descriptive statistics, modelling, and model evaluation. The software also offers
98 a range of add-ons that provide additional functionality for advanced ML tasks, such as
99 computer vision and natural language processing. A significant benefit of using *Orange* is
100 that it is free to download and use across multiple operating systems. As *Orange* is a desktop

application, it can also be used offline without an internet connection. We selected *Orange* as the tool for the workshop for these reasons, so that participants were not limited by the cost of software or requiring a particular operating system.

This paper highlights and reflects on the workshop's design, implementation and delivery, outlining the successes, challenges, and future directions. This paper contributes to the literature by providing a blueprint for teaching domain-specific computational literacy in a way that is time-efficient, transparent, and accessible for participants. This workshop also laid the foundation for the practical *Orange*-based sessions delivered by BM at the recent ESCMID 4-day artificial intelligence workshop [5].

6. Rationale and motivation

6.1 The skills gap in infection science

Infection science professionals routinely generate and interact with large volumes of digital data, such as culture and antimicrobial susceptibility testing results, clinical patient information, test turnaround times, and operational workflow metrics. Despite this, structured training in data analytics is often limited. Traditional laboratory education emphasises diagnostic reasoning and quality assurance but does not necessarily address statistical computing, ML, and AI. Whilst the potential of analytics may be recognised, the inability to translate the analytical requirements into practice limits progress. These one-day workshops have been designed to address this educational gap by introducing the ML lifecycle using an open-source tool that requires no prior programming or advanced computational experience.

6.2 Motivation

The workshop is designed in response to three pedagogical needs identified through previous experience by authors BM and TI. These include the following:

Accessibility: Provide an entry point to data science and ML for participants without prior computational experience.

Relevance: Contextualise examples around infection-science scenarios to make abstract concepts meaningful. This includes focusing on tabular datasets, as this is the data modality participants are most likely to use on a regular basis.

Transparency: Promote transparent, open, and reproducible data science practices, encouraging participants to document analytical steps and adopt open-source tools where possible.

These principles align with a broader emerging philosophy regarding the importance of responsible and trustworthy data science and ML implementation [6,7,8, 9]. By focusing on open workflow design, participants learn how to make their analyses interpretable, shareable, and auditable. These aspects of the analytical workflow are of fundamental importance to clinical, laboratory, and research settings since they form the foundation for the one-day workshops.

7. Workshop design and framework

7.1 Design principles

The workshop is designed to ensure that the intended learning outcomes and teaching activities are explicitly aligned [10]. Noting that assessment or evaluation tasks for the participants are not included. Learning outcomes are clearly articulated at the outset of the workshop, and participants are informed of the knowledge and competencies they are expected to develop by the end of the session. Teaching activities are deliberately structured to support these outcomes, with an emphasis on conceptual understanding of the ML lifecycle rather than narrow technical mastery.

In addition to constructive alignment, the workshop draws on principles from experiential learning theory and cognitive apprenticeship. Experiential learning theory emphasises learning through a cycle of concrete experience, reflective observation, abstract conceptualisation, and active experimentation [11]. This is operationalised through live demonstrations of analytical workflows, hands-on sessions, and opportunities for reflection on methodological decisions. Cognitive apprenticeship further informed the workshop design by recognising that analytical proficiency develops through observation of expert practice, guided participation, and gradual transfer of knowledge and skills to the participants [12]. Guided by this educational framework and the needs of a multidisciplinary audience, the workshop was designed around the following principles:

Authenticity: The analytical techniques and workflows taught during the workshop reflect those commonly required in real-world projects involving clinical and laboratory datasets in tabular/structured form. By focusing on broadly applicable methods rather than highly specialised implementations, participants are encouraged to directly map workshop content to their own research or professional practice.

Progressive scaffolding: The workshop follows a linear and cumulative structure, with each session explicitly building on concepts and skills introduced in prior sessions. This progressive scaffolding supports participants with varying levels of prior experience, enabling them to incrementally develop competence through guided practice.

Reproducibility and transparency: The workshops are grounded in the principles of the scientific method, with reproducibility positioned as a central priority. Participants are introduced to best practices for structuring analytical workflows, documenting experimental decisions, and reporting results in a transparent and trustworthy way.

Experimentation and exploration: Participants are encouraged to actively experiment with the techniques introduced during the workshop, including applying them to their own datasets where feasible. This emphasis on exploration reflects experiential learning principles, reinforcing the iterative nature of ML and supporting deeper engagement with both methodological choices and their practical implications.

Together, these design principles support the development of ML literacy by prioritising conceptual understanding, critical thinking, and practical relevance over exhaustive technical detail, while remaining aligned with established theories of learning and instruction.

7.2 Workshop structure

The one-day workshops are structured to provide participants with fundamental background knowledge in ML and data science, before spending the remainder of the workshop focusing on the practical implementation and expression of these fundamentals. Collectively, we refer to these practical components as “hands-on sessions”. The Original workshop in 2019 designed and delivered by TI was the foundation for the design of the subsequent workshops. The major difference between them is one of scope. In 2019, the focus of the workshop was on applying *Orange* specifically to antimicrobial susceptibility testing data. Following workshops designed by BM and TI focused on working with tabular datasets more generally, as these are the types of datasets that practitioners are more likely to encounter in their own environments. Most practitioners would also be familiar with tabular datasets. The timeline for each of the workshops, along with the topic of each session is presented in Table 1.

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2019 ECCMID (time: topic)	2024 ECCMID (time: topic)	2025 ESCMID Global (time: topic)	2025 Universit y of Zurich (time: topic)	2026 ESCMID Global (time: topic)	2026 University of Zurich (time: topic)
09:30: Welcome	09:00: Welcome, Data science and machine learning backgroun d	09:00: Welcome, Data science and machine learning backgroun d	09:00: Welcome , Data science and machine learning backgrou nd	9:00 – 9:30: Workshop welcome/introdu ction to data science and machine learning	9:00 – 9:30: Workshop welcome/introdu ction to data science and machine learning
09:35: Broad principles of machine learning	10:00 – 10:45: Hands on session 1 – Introducti on to Orange	10:00 – 10:45: Hands on session 1 – Introducti on to Orange	10:00 – 10:45: Hands on session 1 – Introduct ion to Orange	9:30 – 10:00: Understanding the “end-to-end machine learning lifecycle” and hands on session 1– Introduction to Orange	9:30 – 10:00: Understanding the “end-to-end machine learning lifecycle” and hands on session 1 – Introduction to Orange
10:00: Introducing open-source machine learning software	10:45 – 11:15: Morning break	10:45 – 11:15: Morning break	10:45 – 11:15: Morning break	10:00 – 10:45: Hands on session 2 - Data cleaning and preparation	10:00 – 10:45: Hands on session 2 - Data cleaning and preparation
10:15: Step 1: Caring for your data –	11:15 – 12:00:	11:15 – 12:00:	11:15 – 12:00: Hands on	10:45 – 11:15: Morning break	10:45 – 11:15: Morning break

scraping and cleaning	Hands on session 2 - Data cleaning and preparation	Hands on session 2 - Data cleaning and preparation	session 2 - Data cleaning and preparation		
10:45: Morning break	12:00 – 13:00: Hands on session 3 - Training machine learning models	12:00 – 13:00: Hands on session 3 - Training machine learning models	12:00 – 13:00: Hands on session 3 - Training machine learning models	11:15 – 12:00: Hands on session 3 - Training machine learning models	11:15 – 12:00: Hands on session 3 - Training machine learning models
11:15: Step 2: Define your control population with data machine 1	13:00 – 14:00: Lunch break	13:00 – 14:00: Lunch break	13:00 – 14:00: Lunch break	12:00 – 13:00: Hands on session 4 - Evaluating machine learning models	12:00 – 13:00: Hands on session 4 - Evaluating machine learning models
12:00: Step 3: First look AST with data machine 2	14:00 – 15:00: Hands on session 4 - Evaluating machine learning models	14:00 – 15:00: Hands on session 4 - Evaluating machine learning models	14:00 – 15:00: Hands on session 4 and 5 - Evaluating machine learning models and testing our models with new data and	13:00 – 14:00: Lunch break	13:00 – 14:00: Lunch break

			retraining our model		
13:00: Lunch break and troubleshooting	15:00 – 16:00: Hands on session 5 - Testing our models with new data and retraining our model	15:00 – 16:00: Hands on session 5 - Testing our models with new data and retraining our model	15:00 – 16:00: Hands on session 6- More machine learning workflow examples	14:00 – 15:00: Hands on session 5 - Testing our models with new data and retraining our model	14:00 – 15:00: Hands on session 5 - Testing our models with new data and retraining our model Hands on session 6 - More machine learning workflow examples
14:00: Step 4: Closer look at AST with data machine 3	16:00 – 17:00: Hands on session 6 - More machine learning workflow examples	16:00 – 16:30: hands on session 6- More machine learning workflow examples	16:00: Question s, technical assistance, and discussion.	15:00 – 16:00: Hands on session 6 - More machine learning workflow examples	15:00 – 16:00: Guidance for publishing machine learning research
15:00 Step 5: Create your data machine ensemble	17:00 – 18:00: Close workshop - Opportunities for further questions, technical	16:30 – 17:00: Presentation by Tim Inglis regarding application of data science and		16:00 – 17:00: Guidance for publishing machine learning research and presentation by Tim Inglis regarding application of data science and	16:00 – 17:00: Close workshop - Opportunities for further questions, technical assistance, networking

	assistance, networkin g	machine learning for infection science and clinical microbiolo gy.		machine learning for infection science and clinical microbiology	
15:30: Summative exercise		17:00 – 18:00: Close workshop - Opportuni ties for further questions, technical assistance, networkin g		17:00 – 18:00: Close workshop - Opportunities for further questions, technical assistance, networking	
16:00: Review					

Table 1: The timeline and topic of each session for the workshops presented in 2019,2024, 2025, and 2026

8. Workshop implementation and delivery

8.1 Workshop preparation

The workshops are taught using the *Orange* data mining software. To ensure that the workshop is delivered efficiently, it is critical that participants are appropriately set up with *Orange* on their own laptops before attending the workshop. To coordinate this, BM developed a brief instructional document to guide participants in installing *Orange* on their own laptops. The instruction documents have been developed for Windows, Mac, and Linux (Ubuntu) operating systems. They are sent out to all participants as part of the pre-workshop communications strategy. This strategy, which ensures that participants are prepared for the workshop, is outlined in Table 2. Across all workshops, participants were appropriately set up when they arrived, demonstrating the effectiveness of this communication approach.

The course material is also provided to participants, made available to them via a link to a password protected *Dropbox* folder. The content remains available to participants beyond the conclusion of the workshop.

The rooms used for the workshops are of a classroom style set up, with access to power at each individual desk so that participants can ensure their devices have sufficient battery power to sustain the activities throughout the duration of the workshop.

Weeks until workshop	Email content
8	- Facilitator introduction. - Summary of the content in the workshop. - Instructions for installing <i>Orange</i>.
4	- Sharing the Dropbox link that contains all the workshop material. - Sharing the schedule for the workshop. - Instructions for participants to bring their own datasets if they are interested.
1	- Reminder of start time of the workshop and details regarding venue. - Reminders regarding accessing the content and downloading the <i>Orange</i> software.

Table 2: Due to the international nature of the workshop, email is used to communicate with participants. Emails are sent 8, 4, and 1 week before the workshop, with each email containing content to support appropriate participant preparation

8.2 Workshop delivery

The workshop content is typically taught by both BM and TI. BM delivers the session content, both theoretical and hands-on sessions, with TI providing technical support and clinical microbiology expertise. Having a professional data scientist (BM) and a clinical microbiologist (TI) delivering the workshop allows for a more diverse range of skill sets and knowledge to support participants with their questions and inquiries. Both BM and TI acted as coaches during the hands-on sessions, modelling the content to participants and demonstrating reasoning and transparent problem-solving practices. Participants are encouraged to ask questions and participate throughout the hands-on sessions. The break periods, as shown in Table 1, provide opportunities for students to consolidate what they

have learnt, ask questions, receive technical support, or have a rest before starting the next set of sessions.

8.3 Session content

8.3.1 Introductory presentation

The workshops start with a one-hour presentation. This presentation consists of a general introduction to the workshop and outlines the learning goals. Participants are shown the final *Orange* workflow that they build by the conclusion of the workshop (Figure 1). The workflow that participants build is based on previous work by BM, TI, and MR, regarding ML for blood culture outcome prediction [13]. To the best of our knowledge, this is the first time that a workshop has been delivered that presents a workflow for blood culture outcome prediction. However, the workflows can be easily adapted, and participants are encouraged to bring their own datasets to the workshop and apply what they learn to their specific problems. Following this introduction, for the remaining hour of the first session, we cover fundamental data science, and ML concepts, and provide some examples of where ML has been applied to infectious diseases and clinical microbiology. Following this, we introduce the data science and ML lifecycle to participants, outlining the main components, including 1) problem identification and ML objective; 2) data preparation; 3) model development; and 4) evaluation. We then provide additional background on structured data and supervised ML.

{{TYPE:image; FILENAME:image1.png; ID:rld13}}

Figure 1: Final analytical workflow developed by the participants of the workshop. Includes data cleaning and processing, training and testing machine learning models, cross-validation, and model evaluation.

8.3.2 Hands-on session 1

The first hands-on session is designed to introduce participants to the *Orange* software ecosystem in preparation for the remainder of the workshop. The session utilises publicly available, published data [14], and its focus is on how to use *Orange*. This includes importing data into the application and visualising it. The resulting workflow produced during this hands-on session is shown in Figure 2.

{{TYPE:image; FILENAME:image2.png; ID:rld14}}

Figure 2: Workflow that participants build during the first hands-on session. Participants learn how to import data, visualise it in a table, and perform exploratory data analysis. The purpose of this session is to show participants how to use the Orange software.

8.3.3 Hands-on session 2

Following the first break, we start with hands-on session 2. This is the first hands-on session that involves the BC outcome prediction use case. The focus of this session is data cleaning and preparation for ML modelling. The session starts with importing the data, followed by participants learning how to filter data, create new columns/features, and select columns/features, before concluding with the training and test data split. The focus of this session is for participants to understand how to prepare tabular datasets for ML and highlight the importance of the training and test split. The workflow produced during this session is shown in Figure 3.

{{TYPE:image; FILENAME:image3.png; ID:rld15}}

Figure 3: In hands-on session 2, participants focus on data cleaning and preparation for machine learning. Starting with importing the data and concluding with the creation of training and test datasets.

8.3.4 Hands-on session 3

Hands-on session 3 introduces the ML training and cross-validation process. We discuss the importance of cross-validation and why it is a critical component of any ML pipeline. For the modelling, we start with tree-based models and explain the importance of model hyperparameters. As hands-on session 2 and hands-on session 3 are both in-depth sessions, they are followed by a one-hour break. The workflow produced by the end of hands-on session 3 is shown in Figure 4.

{{TYPE:image; FILENAME:image4.png; ID:rld16}}

Figure 4: In hands-on session 3, participants focus on the training of machine learning models, building on the previous session.

8.3.5 Hands-on session 4

Hands-on session 4 focuses on the evaluation of ML models. An entire session is dedicated to this topic, as it is very important for those aiming to develop their own ML models and those reviewing ML models developed by others. Performance metrics can be misleading in

the context of highly imbalanced datasets, such as the one participants use in the workshops, and this is discussed in detail. In this session, we primarily focus on the area under the receiver operating characteristic curve (AUC), and metrics relating to the confusion matrix such as sensitivity and specificity. We also build on the previous session to go beyond training and cross-validation performance and examine model performance on the test set we prepared earlier. The workflow produced by the end of this session is shown in Figure 5.

{{TYPE:image; FILENAME:image5.png; ID:rld17}}

Figure 5: Hands-on session 4 focuses on evaluating the models trained in the previous session. This includes evaluating the models during cross-validation and assessing their performance on the test set.

8.3.6 Hands-on session 5

Hands-on session 5 focuses on the example case of using an external validation set and how participants can adapt their workflow to accommodate this. By the end of this session, participants have completed the entire BC outcome prediction workflow (Figure 1). Likewise, participants using their own datasets should have a similar workflow adapted to their particular purpose.

8.3.7 Hands-on session 6

Hands-on session 6 extends on what participants have learnt over the course of the workshop and introduces some different types of workflows. These include workflows where more data cleaning is required, such as missing value handling and encoding of categorical variables. Multiclass classification is also discussed. Additional workflows for regression tasks, unsupervised learning, and computer vision are also introduced.

All participants receive the workshop materials before the workshop, and the content remains available to them after it concludes. Participants are provided with a record of all email communication, instructions for installing the *Orange* software, the workshop presentation slides, copies of the completed *Orange* workflows, and additional supplementary material, including the datasets.

9. Reflections

After delivering this workshop on multiple occasions to a diverse audience, there is a clear opportunity for reflection. Participants have come from a wide range of backgrounds, including medical microbiology, infectious diseases, laboratory science, epidemiology, data science, and graduate programmes. Designing and delivering a workshop that is valuable across such disciplinary diversity is challenging. This diversity is a key motivator for presenting the material in the most generalisable way possible, with an emphasis on shared concepts, workflows, and decision-making processes rather than on discipline-specific technical detail.

Levels of engagement varied across the range of participants, reflecting both differences in prior experience and the intensity of the material. A recurring challenge is the risk of information overload. While the one-day format is beneficial in enabling efficient content delivery, it is also inherently constrained. There is a practical limit to the depth achievable in a single day, and depth is often sacrificed in favour of conceptual coherence. The primary objective of the workshop, therefore, is not complete technical mastery, but rather for participants to leave with robust mental models of the ML lifecycle, alongside introductory technical skills and improved literacy.

This trade-off is an acknowledged limitation of the one-day format. The contrast became particularly evident during delivery of a multi-day version of the workshop, in which content could be distributed across several days [5]. In this setting, concepts were reinforced iteratively, allowing participants more time to reflect, consolidate learning, and build progressively on prior material. Additionally, the workshop is delivered exclusively in English, which may limit accessibility for participants whose primary language is not English.

10. Future directions

Across all workshops delivered in 2024, 2025, and 2026, the material has been taught to 220 participants from a range of countries and professional backgrounds. There are plans to continue offering this workshop in future years if demand remains. Beyond in-person delivery, a key future objective is to ensure that the workshop material is accessible to the wider infection science community. In support of this goal, an open-source repository will be created to host the workshop and supplementary content. The repository will serve as a freely available, regularly updated educational resource for those interested in applying machine learning to infection science. Its open-source nature will enable community contribution, adaptation, and extension, including a roadmap for future development.

Participant feedback was collected following workshop delivery and provided valuable insights into potential improvements. However, ethics approval was not obtained to formally

analyse or report this feedback; therefore, it is not included in this paper. Future research ethics-approved work may enable systematic evaluation of participant feedback and its use in future reports.

While quantitative and qualitative feedback is important for evaluating the success of educational initiatives, broader indicators of impact should also be considered. Reflections, such as whether participants became more critical consumers of ML literature or whether new collaborations and projects emerged following the workshop, may provide more meaningful insight into the workshop's achievement of its intended objectives.

Although the one-day workshop format is efficient and effective practical introduction to the ML lifecycle, it does not scale infinitely. There is a finite amount of material that can be reasonably accommodated within the current structure. Multi-day formats are better suited to spaced repetition, deeper exploration of complex topics, and more sustained engagement. Expanding the delivery format represents an important area for future development. For example, the workshop could be delivered online across multiple shorter sessions, improving accessibility and reducing resource requirements associated with in-person attendance.

Finally, given the multidisciplinary nature of the participant cohort, the workshop provides a valuable opportunity for networking and collaboration. Future iterations should place greater emphasis on facilitating interaction and collaboration among participants as a core component of the workshop design. We hope that this paper will encourage other educators to develop similar workshops based on the framework presented here, thereby supporting broader efforts to improve ML literacy within the infection science community.

11. Conclusion

The *"Using Data Science and Machine Learning for Infection Science: A Hands-On Introduction"* workshops demonstrate how carefully designed, one-day educational interventions can support development of computational literacy among infection science professionals, while maintaining a commitment to transparency and open science. By integrating foundational theory of the ML lifecycle with real-time demonstrations and hands-on sessions, the workshop offers a replicable model for domain-specific data science education.

Successful delivery of the workshop on multiple occasions suggests that this approach is both feasible and valuable for the infection science and clinical microbiology community. As the workshop continues to evolve, ongoing reflection is needed to identify opportunities for improvement, expansion, and broader dissemination of the material. By sharing both the content and lessons learned through open channels, we hope that this work will become a

blueprint for similar educational initiatives and contribute to more informed, critical, and responsible use of ML in infection science and clinical microbiology.

12. Author statements

12.1 Author contributions

BM wrote the initial draft of this manuscript and designed and delivered the workshops in 2024, 2025, and 2026. TI reviewed the manuscript, provided editorial support, and supervision. TI designed and delivered the 2019 workshop, mentioned in this paper. TI was also present at the ECCMID and ESCMID Global workshops in 2024, 2025, and 2026 to provide technical assistance for participants. MR provided editorial support.

12.2 Conflicts of interest

The author(s) declare that there are no conflicts of interest.

12.3 Funding information

For the workshops in Zurich, travel and accommodation costs were covered by the University of Zurich. BM and TI received waivers of conference registration fees at ECCMID/ESCMID Global.

12.4 Ethical approval

Not applicable

12.5 Consent for publication

Not applicable

12.6 Acknowledgements

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